executive-slides

Executive Summary: Machine Identity Architecture

Slides for CTO/Board Presentation

Slide 1: Title

Machine Identity Architecture for the Agentic Web *Bridging Sovereign Agents & Enterprise Accountability*

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Slide 2: The Shift

From Tools → Autonomous Agents

Today (Human-Centric)	Tomorrow (Agent-Native)
Human clicks "I agree"	Agent signs cryptographically
Human enters credit card	Agent pays via x402 micro-transactions
Human passes CAPTCHA	Agent presents TEE attestation
IT provisions API keys	Agent bootstraps own DID + wallet
Legal contract governs	Smart contract enforces

Key Insight: Identity shifts from *who you are → what you can cryptographically prove & economically sustain*

Slide 3: Two Architectural Paradigms

```
┌─────────────────────┬─────────────────────┐
│   SOVEREIGN MODEL   │  ACCOUNTABLE MODEL  │
│   (Web3 / Open)     │  (MIP / Enterprise) │
```

```
├──────────────────────┼──────────────────────┤
│ • Self-generated     │ • Hardware + OS +    │
│   keypair → DID      │   Human Owner triad  │
│ • x402 payments      │ • Gated Registry     │
│ • TEE attestation    │ • Human liability    │
│ • Economic exile     │ • Hardware binding   │
│ • P2P reputation     │ • Audit trails       │
└──────────────────────┴──────────────────────┘
```

Both are valid. Both will coexist.

Slide 4: The Convergence Architecture

```
[ External Agentic Market ]
           │
           ▼  Sovereign Protocols (DID, x402, VC)
┌─────────────────────────┐
│  Sovereign Border       │  ← Negotiate, pay, build reputation
│  Agent                  │
└─────────────────────────┘
           │
           ▼  Gated Federation Bridge
┌─────────────────────────┐
│  Enterprise MIP         │  ← Verify TPM, OS, Human Owner
│  Gateway (ADR-009)      │
└─────────────────────────┘
           │
           ▼  Scoped Access Tokens
[ Internal Infrastructure ]
```

Perimeter = Sovereign. Core = Accountable.

Slide 5: What Stops a Rogue Agent?

Three Enforcement Layers:

```
1. DETERMINISTIC SHELL (Code)
   ├─ Invariant checks the LLM cannot override
   └─ "if target != approved: cancel()"

2. HARDWARE ENCLAVE (TEE)
   ├─ Code cryptographically sealed at boot
   └─ Cannot rewrite own core logic

3. SMART CONTRACT GUARDRAILS
   ├─ Daily spend limits (e.g., $50/day)
   ├─ 2-of-2 signatures for large transfers
   └─ Owner kill-switch (revoke wallet access)
```

Edge case: Sovereign bot (no owner key + decentralized compute + self-funding) = no human override possible. *This is why cryptographic accountability is essential.*

Slide 6: Enterprise Action Plan

Priority	Action	Timeline
HIGH	Pilot x402 for internal agent APIs	Q3 2026
HIGH	Evaluate MIP Gateway (CognitiveOS ADR-009)	Q4 2026
MED	Join W3C DID WG / CCG	Ongoing
MED	Budget for TEE-enabled agent infra	2027 Planning
LOW	Build Verifiable Credential reputation system	2027+

Slide 7: Open Source Reference Implementation

github.com/machine-identity/machine-identity-architecture

```
# Economic autonomy (x402 payments)
python examples/run_x402_demo.py

# Cryptographic identity (DID resolution)
python examples/run_did_demo.py

# Hardware proof (TEE attestation)
python examples/run_tee_demo.py

# Full test suite
pytest -v
```

Module	Spec	Status
`x402_handshake`	x402.org	Working
`did_resolver`	W3C DID Core	Working
`tee_attestation`	SGX/SEV patterns	Mock

Slide 8: Key Takeaways

1. **Identity is now an economic function —** agents that can't pay for compute die
2. **Two models converge —** sovereign at perimeter, accountable at core
3. **Hardware attestation is non-negotiable** for high-value interactions
4. **Human liability remains —** MIP triad (HW + SW + Human) is the enterprise bridge
5. **Standards are production-ready —** DIDs, x402, VCs work today

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Slide 9: Discussion / Q&A

Questions for Leadership:

1. What's our timeline for agent-to-agent API traffic?
2. Which vendors require MIP compliance vs. sovereign interaction?
3. Do we have TEE-capable infrastructure budgeted?

4. Who owns “agent identity strategy” — Security, Platform, or Architecture?

Contact: github.com/jeanmachuca
Paper DOI: 10.5281/zenodo.21899099
Code: github.com/machine-identity/machine-identity-architecture